Research Notes on Question answering

Question answering

>> Section 1

Fact-checking if a fact is verified, by posing a question like: is fact X true or false? Customer service, Technical support, Market research, Natural language generation of reports or research.

>> Section 2

Mathematics An open-source, math-aware, question answering system called MathQA, based on Ask Platypus and Wikidata, was published in 2018. MathQA takes an English or Hindi natural language question as input and returns a mathematical formula retrieved from Wikidata as a succinct answer, translated into a computable form that allows the user to insert values for the variables. The system retrieves names and values of variables and common constants from Wikidata if those are available. It is claimed that the system outperforms a commercial computational mathematical knowledge engine on a test set. MathQA is hosted by Wikimedia at https://mathqa.wmflabs.org/. In 2022, it was extended to answer 15 math question types. MathQA methods need to combine natural and formula language. One possible approach is to perform supervised annotation via Entity Linking. The "ARQMath Task" at CLEF 2020 was launched to address the problem of linking newly posted questions from the platform Math Stack Exchange to existing ones that were already answered by the community. Providing hyperlinks to already answered, semantically related questions helps users to get answers earlier but is a challenging problem because semantic relatedness is not trivial. The lab was motivated by the fact that 20% of mathematical queries in general-purpose search engines are expressed as well-formed questions. The challenge contained two separate sub-tasks. Task 1: "Answer retrieval" matching old post answers to newly posed questions, and Task 2: "Formula retrieval" matching old post formulae to new questions. Starting with the domain of mathematics, which involves formula language, the goal is to later extend the task to other domains (e.g., STEM disciplines, such as chemistry, biology, etc.), which employ other types of special notation (e.g., chemical formulae). The inverse of mathematical question answering—mathematical question generation—has also been researched. The PhysWikiQuiz physics question generation and test engine retrieves mathematical formulae from Wikidata together with semantic information about their constituting identifiers (names and values of variables). The formulae are then rearranged to generate a set of formula variants. Subsequently, the variables are substituted with random values to generate a large number of different questions suitable for individual student tests.

>> Section 3

Further reading Dragomir R. Radev, John Prager, and Valerie Samn. Ranking suspected answers to natural language questions using predictive annotation Archived 2011-08-26 at the Wayback Machine. In Proceedings of the 6th Conference on Applied Natural Language Processing, Seattle, WA, May 2000. John Prager, Eric Brown, Anni Coden, and Dragomir Radev. Question-answering by predictive annotation Archived 2011-08-23 at the Wayback Machine. In Proceedings, 23rd Annual International ACM SIGIR Conference on Research and Development in Information Retrieval, Athens, Greece, July 2000. Hutchins, W. John; Harold L. Somers (1992). An Introduction to Machine Translation. London: Academic Press. ISBN 978-0-12-362830-5. L. Fortnow, Steve Homer (2002/2003). A Short History of Computational Complexity. In D. van Dalen, J. Dawson, and A. Kanamori, editors, The History of Mathematical Logic. North-Holland, Amsterdam. Tunstall, Lewis (5 July 2022). Natural Language Processing with Transformers: Building Language Applications with Hugging Face (2nd ed.). O'Reilly UK Ltd. p. Chapter 7. ISBN 978-1098136796.

>> Section 4

In information retrieval, an open-domain question answering system tries to return an answer in response to the user's question. The returned answer is in the form of short texts rather than a list of relevant documents. The system finds answers by using a combination of techniques from computational linguistics, information retrieval, and knowledge representation. The system takes a natural language question as an input rather than a set of keywords, for example: "When is the national day of China?" It then transforms this input sentence into a query in its logical form. Accepting natural language questions makes the system more user-friendly, but harder to implement, as there are a variety of question types and the system will have to identify the correct one in order to give a sensible answer. Assigning a question type to the question is a crucial task; the entire answer extraction process relies on finding the correct question type and hence the correct answer type. Keyword extraction is the first step in identifying the input question type. In some cases, words clearly indicate the question type, e.g., "Who", "Where", "When", or "How many"—these words might suggest to the system that the answers should be of type "Person", "Location", "Date", or "Number", respectively. POS (part-of-speech) tagging and syntactic parsing techniques can also determine the answer type. In the example above, the subject is "Chinese National Day", the predicate is "is" and the adverbial modifier is "when", therefore the answer type is "Date". Unfortunately, some interrogative words like "Which", "What", or "How" do not correspond to unambiguous answer types: Each can represent more than one type. In situations like this, other words in the question need to be considered. A lexical dictionary such as WordNet can be used for understanding the context. Once the system identifies the question type, it uses an information retrieval system to find a set of documents that contain the correct keywords. A tagger and NP/Verb Group chunker can verify whether the correct entities and relations are mentioned in the found documents. For questions such as "Who" or "Where", a named-entity recogniser finds relevant "Person" and "Location" names from the retrieved documents. Only the relevant paragraphs are selected for ranking. A vector space model can classify the candidate answers. Check if the answer is of the correct type as determined in the question type analysis stage. An inference technique can validate the candidate answers. A score is then given to each of these candidates according to the number of question words it contains and how close these words are to the candidate—the more and the closer the better. The answer is then translated by parsing into a compact and meaningful representation. In the previous example, the expected output answer is "1st Oct."

>> Section 5

Question answering (QA) is a computer science discipline within the fields of information retrieval and natural language processing (NLP) that is concerned with building systems that automatically answer questions that are posed by humans in a natural language. A question-answering implementation, usually a computer program, may construct its answers by querying a structured database of knowledge or information, usually a knowledge base. More commonly, question-answering systems can pull answers from an unstructured collection of natural language documents. Some examples of natural language document collections used for question answering systems include reference texts, compiled newswire reports, Wikipedia pages and other World Wide Web pages.